Experiment No. 02

Title: Software Project Management Plan (SPMP)

Roll No.: 16014225801 **Batch: C3** **Experiment No.:02**

Aim: To Prepare Software Project Management Plan (SPMP)

Resources needed: Internet Explorer, GanttProject, LaTeX Editor

Pre Requisite: Knowledge of software engineering fundamentals, software process models, and project scheduling concepts.

Theory:

The software project management plan specifies:

- the objectives of the project,
- the project dependencies and constraints,
- the project deliverables, and
- the project timetable.

This document describes how you intend to develop your software; how you expect to approach the development process; the times and deliverables involved; and the problems foreseen and constraints imposed. The plan accompanies the project life span and is usually subject to a number of revisions.

SPMP Template:

1 INTRODUCTION

1.1 Project Overview

Provide a description of the purpose of the softwaxre, who the project is being executed for and the expected delivery date.

1.2 Project Deliverables

List the project deliverables / "work products" (documents, source code, library files, executable code, databases), and the delivery date for each (the date that each is to be complete).

2 PROJECT ORGANIZATION

2.1 Software Process Model

Identify and describe the software process model you will use in terms of the following:

- the flow of information and work products,
- reviews to be conducted,
- major milestones to be achieved,
- versions to be established,
- project deliverables to be completed.

To describe the process model, a combination of graphical and textual notations may be used.

2.2 Roles and Responsibilities

In the case of a group project, identify the various roles, the project team members and their assignments. Take care to identify how information will be communicated between the

different roles — for example, how the design will be passed to the coder. Use diagrams to show the project structure and the lines of communication within the project.

2.3 Tools and Techniques

Specify the development methodologies, notations, programming languages, techniques and tools you plan to use at each stage of the project. For example, you might use OO for the analysis phase, using the UML notation and the Rational Rose tool; to track changes to the analysis you might use version control and the CS-RCS tool. This section should include details of the coding standards to be used.

3 PROJECT MANAGEMENT PLAN

3.1 Tasks

Identify the tasks involved in executing the project. Some sample tasks are:

- requirements analysis / clarify the requirements
- requirements analysis / develop an exploratory prototype and get feedback from the user
- requirements analysis / write the SRS
- system design / develop the design using Rational Rose
- system design / write the SDD
- development / develop version 1
- development / develop version 2
- system test / test version 1
- system test / test version 2

Tasks should be specified in enough detail to allow estimation of the time required and to allow tracking of project status. As a very rough guide, a task should take about one week.

3.1.n Task-n

Provide a name for the task and a unique identifier of the form SPMP-Tnnnn.

3.1.n.1 Description

A brief description of the task.

3.1.n.2 Deliverables and Milestones

A list of the deliverables of the task (documentation, code, other) and the milestones associated with these. For example, milestone 1 might be delivery of the final revisions of the SRS, STD and SPMP; milestone 2 might be delivery of the STD; milestone 3 might be completion of version 1 (development and test), etc.

3.1.n.3 Resources Needed

Identify the resources needed to execute the task (equipment, access to papers / books, etc.).

3.1.n.4 Dependencies and Constraints

Identify the dependencies that must be met before the task can be started or completed, and any constraints placed on the task. For example, detailed design cannot start until the requirements are complete, or there is a constraint that Java be used as the programming language.

3.1.n.5 Risks and Contingencies

Identify the risks associated with getting this task completed on time, and the contingency plans to cope with each risk. You can never identify all the risks, and some events would stop the project completely, so limit this section to reasonable risks. For example, what is the

probability of being unable to get the programming language running on your computer, and what would you do to recover?

3.2 Assignments

For group projects, identify the assignment of team members to tasks.

3.3 Timetable

Provide a timetable showing the estimated start and completion dates for each task. This is best illustrated using a Gantt chart diagram. This timetable should be updated throughout the project to reflect changes in the tasks, the execution time for tasks, the completion dates, etc.

Procedure:

1. Prepare a Gantt chart using GanttProject.

2. Prepare the SPMP document for the chosen problem definition in LaTeX.

Results:

Attached

Questions:

1. Can a project be undertaken without a plan? What are the possible consequences?

2. What are the key components of a Software Project Management Plan (SPMP)?

3. What is a Gantt chart and how is it useful in project scheduling?

4. Differentiate between a milestone and a deliverable in a software project.

5. Why is a project timetable subject to a number of revisions during the project life span?

Answers:

1. No, a project should not be undertaken without a proper plan. Without planning, the project may face unclear objectives, poor resource allocation, delays, cost overruns, confusion among team members, increased risks, poor quality, and possible project failure.

2. The key components are:
- Project overview and objectives
- Project scope
- Project organization and responsibilities
- Project schedule and milestones
- Resource and cost management
- Risk management
- Quality management
- Communication plan
- Configuration and change management
- Project monitoring and control

3. A Gantt chart is a graphical representation of project activities against a timeline. It shows task duration, start and finish dates, dependencies, milestones, and progress. It helps managers schedule tasks, allocate resources, monitor progress, and identify delays.

4. A milestone is a significant event or achievement in a project that indicates the completion of an important stage. It normally has no duration. A deliverable is a tangible or measurable output produced during the project, such as an SRS document, software module, test report, or final software product.

5. A project timetable is revised because of changes in requirements, unexpected technical problems, inaccurate initial estimates, resource changes, task delays, new risks, changes in project scope, and dependencies between activities. Therefore, the timetable must be updated according to the actual progress of the project.

Conclusion:
The SPMP was successfully prepared, defining project objectives, deliverables, responsibilities, tasks, risks, resources, and timetable for effective project management.

Grade: AA / AB / BB / BC / CC / CD /DD

Signature of faculty in-charge with date

References:
Books:
1. Roger S. Pressman, Software Engineering: A Practitioner's Approach, 7th Edition, McGraw Hill, 2010.
2. Ian Sommerville, Software Engineering, 9th Edition, Addison Wesley, 2011.
3. John M. Nicholas, Project Management for Business and Technology, 2nd Edition, Pearson Education.